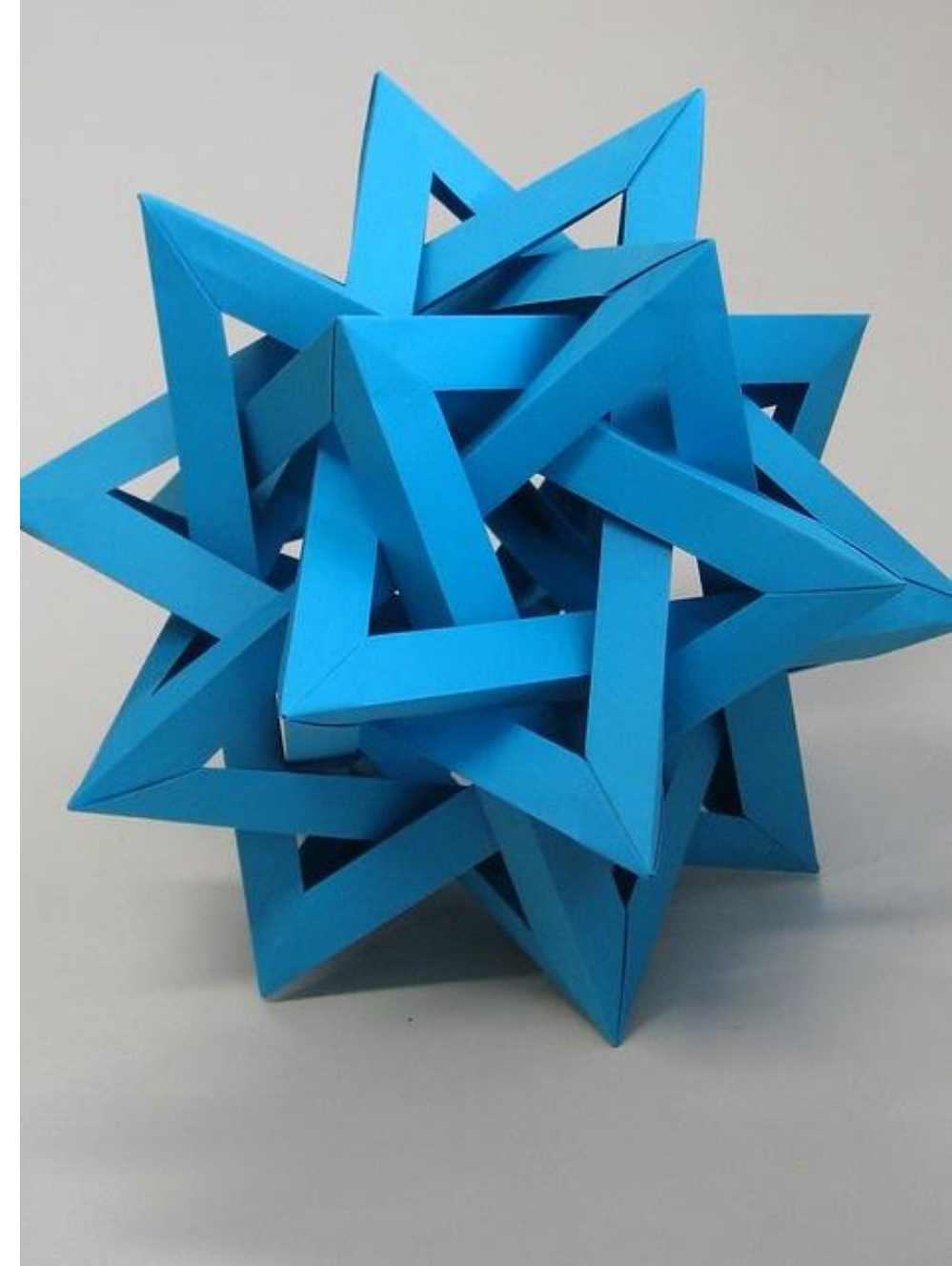




LAB 7

LISTS



Recommended exercises

- **7.1.2**

- **7.1.5**

- **7.2.3**

Exercise 7.1.2

- Read request from Lab text..

Exercise 7.1.2

- Warning: **You are required to print the result in 4 separate lines**
 - Multiple iterations on the list will be needed...
- Remember: we can iterate with
 - `for el in list` (by item)
 - `for idx in range(len( list ))` (by index)
- NOTE: for some cases, you can directly extract the necessary elements from a list using python slicing
 - `List[::3]` for example will take every 3 elements

Exercise 7.1.5

- Read request from Lab text..
- Write the function `def sameSet(a,b)` that tests whether two lists contain the same elements. Ignore the order and any duplicates.
- Example: [1,2,3,3,3] and [1,1,3,1,1,2] will be considered equal by the `sameSet` function

Exercise 7.1.5

- **For each element** of a list, check that the other contains at least one copy of it in **any** position.
- **It must be done for both lists**
 - For each element $x \in \mathbf{a}$ $\rightarrow$ is x also in $\mathbf{b}$?
 - For each element $x \in \mathbf{b}$ $\rightarrow$ is x also in $\mathbf{a}$?
- If not, the lists are to be considered different.
- NOTE: The function returns **True** or **False**

Exercise 7.2.3

- Read request from Lab text..

Exercise 7.2.3

- Game model = list where each element corresponds to 1 pile of cards.
- Two phases
 - 1) Construction of the initial piles
 - 2) Actual game

1) Construction of the initial piles

- Their number is not defined a priori and is determined by the values drawn at random. Example:
 - 1st value (random between 1 and 45) = 12
 - Remaining cards: $45 - 12 = 33$
 - 2nd value (random between 1 and 33) = 8
 - Remaining cards: $33 - 8 = 25$
 - Etc.
- NOTE: The range of `randint()` changes with each iteration.

Exercise 7.2.3

2) Actual game

- Until the final configuration is reached ([1,2,3,4,5,6,7,8,9])
 - Remove a card from each pile and form a new one with the total thus obtained.
 - Remove any empty piles
- WARNING: How to compare the current configuration with the final one? (see Exercise 7.1.5...)